Course 2021 | Semester II | Theory | 4 Credits

Engineering Materials and Measurements

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**4:0:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**6 h/week; 90 h total**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

This handbook is based only on the supplied official **Diploma Curriculum Revision 2026** syllabus PDF for Course **2021**, titled **Engineering Materials and Measurements**. Objectives, course outcomes, teaching scheme, major topics, detailed coverage, activities and assessment are drawn from that source.

Source treatment: PDF table wrapping can interrupt a heading across lines. In such cases, this handbook retains the official intent in a clearly labelled topic. Normal teaching explanations, study flows, diagrams and Malayalam-support boxes are handbook support rather than claims of official wording.

Verified source note: No web research was used to establish syllabus content. Official references and URLs appear only where they are listed in the supplied PDF.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Engineering Materials and Measurements introduces material structure, properties, heat treatment, testing and metrology. It prepares students to relate material choice and dimensional inspection to design, quality assurance and manufacturing practice.

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Prerequisite foundation

Applied Physics for Mechanical, Structural and Industrial Applications (2002A) and Chemistry for Engineering Practices (2003A).

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
4	0	0	0	6 h/week; 90 h total	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Classify materials and explain structure, bonding, crystallography and microstructure.
2. Explain material properties, heat treatment and testing.
3. Use metrology, standards, limits, fits, tolerances and linear/form measurement.
4. Identify angular/geometric measurements, gauges and comparators.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain classification, structure, bonding, crystallography and microstructure of engineering materials.	Understand
CO2	Describe mechanical, thermal, magnetic and electrical properties, heat treatment and testing.	Understand
CO3	Explain metrology, standards, limits, fits, tolerances and linear/form measurements.	Apply
CO4	Explain angular/geometric techniques and identify gauges/comparators for inspection.	Apply

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	Title	Official major topics	Hours	CO	Handbook section
1	Basic Engineering Materials and Structure	Material classification; atomic structure/bonding; crystallography; ferrous/non-ferrous alloys; smart materials and composites.	15 hours	CO1	Module 1
2	Properties, Heat Treatment and Material Testing	Mechanical, thermal, magnetic and electrical properties; heat treatments; metal failure; destructive and non-destructive testing.	15 hours	CO2	Module 2
3	Metrology, Standards, Fits and Form Measurement	Metrology terms and errors; standards/system components; limits/fits/tolerances; vernier/micrometer; geometric form measurements.	15 hours	CO3	Module 3
4	Angular and Geometric Measurement, Gauges and Comparators	Bevel protractor, sine bar, spirit level, dial gauge, depth gauge, clinometer, autocollimator, angle gauges; GO/NO-GO gauges; comparator types.	15 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Basic Engineering Materials and Structure

Material classification; atomic structure/bonding; crystallography; ferrous/non-ferrous alloys; smart materials and composites.

CO1 · 15 hours

Properties, Heat Treatment and Material Testing

Mechanical, thermal, magnetic and electrical properties; heat treatments; metal failure; destructive and non-destructive testing.

CO2 · 15 hours

Metrology, Standards, Fits and Form Measurement

Metrology terms and errors; standards/system components; limits/fits/tolerances; vernier/micrometer; geometric form measurements.

CO3 · 15 hours

Angular and Geometric Measurement, Gauges and Comparators

Bevel protractor, sine bar, spirit level, dial gauge, depth gauge, clinometer, autocollimator, angle gauges; GO/NO-GO gauges; comparator types.

CO4 · 15 hours

MODULE 1

Basic Engineering Materials and Structure

Material classification; atomic structure/bonding; crystallography; ferrous/non-ferrous alloys; smart materials and composites.

15 hours **CO1** **Understand · Apply**

Learning goals

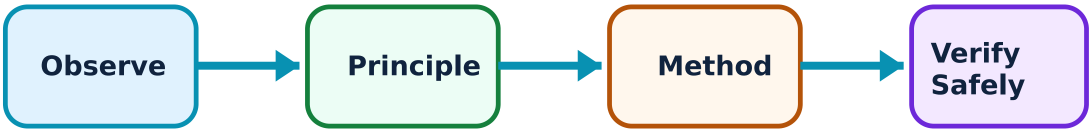
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Basic Engineering Materials and Structure: identify the system, state its principle, follow the correct method and verify the result safely.

1. Material classification and atomic bonding−

Engineering materials are selected by structure, properties, processing and use. Atomic structure and electronic configuration influence bonding. Metallic, ionic and covalent bonding provide different combinations of conductivity, strength and ductility.

Study and application method

Classify a material first, then connect atomic/bonding description to one observable property and application.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Classify a material first, then connect atomic/bonding description to one observable property and application.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Material class does not by itself guarantee performance; service condition and processing also matter.

2. Crystallography and packing−

Crystalline metals have ordered atomic arrangement. Crystal systems, unit-cell parameters, packing factor and BCC/FCC/HCP structures help describe how atoms occupy space and relate structure to properties.

Study and application method

Use a labelled unit-cell sketch and state coordination/packing idea at the required level before comparing structures.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use a labelled unit-cell sketch and state coordination/packing idea at the required level before comparing structures.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not confuse a grain in a real metal with a single unit cell.

3. Ferrous, non-ferrous and special materials−

Steel, stainless steel and cast iron are official ferrous examples. Aluminium/copper alloys, magnesium and titanium alloys are non-ferrous examples. Smart materials and composites are special materials selected for specialised response or combined properties.

Study and application method

Compare composition category, key property and typical engineering use in a table.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare composition category, key property and typical engineering use in a table.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Alloys of a base metal can differ significantly from pure metal; avoid generalising one property.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Properties, Heat Treatment and Material Testing

Mechanical, thermal, magnetic and electrical properties; heat treatments; metal failure; destructive and non-destructive testing.

15 hours | CO2 | Understand · Apply

Learning goals

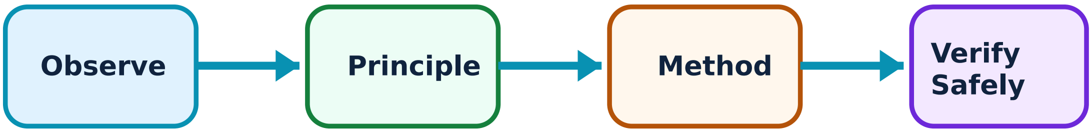
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Properties, Heat Treatment and Material Testing: identify the system, state its principle, follow the correct method and verify the result safely.

1. Mechanical and thermal properties−

Elasticity, plasticity, yield, tensile behaviour, ductility, toughness, hardness, fatigue and creep describe mechanical response. Thermal conductivity and expansion describe heat-related response relevant to service and dimensional change.

Study and application method

Define each property, state a simple test/condition or engineering example, and distinguish strength from hardness/toughness.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define each property, state a simple test/condition or engineering example, and distinguish strength from hardness/toughness.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Hardness, strength and toughness are related but not interchangeable terms.

2. Magnetic and electrical properties−

Permeability, retentivity and coercivity describe magnetic behaviour. Conductivity and resistivity describe electrical transport. Selection depends on whether a component must conduct, resist current, store flux or avoid magnetisation.

Study and application method

State unit/meaning where applicable and link property to a component such as conductor, resistor, core or sensor.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State unit/meaning where applicable and link property to a component such as conductor, resistor, core or sensor.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: High electrical conductivity does not imply high magnetic permeability.

3. Heat treatment, failures and testing−

Annealing, normalizing, hardening, tempering, case hardening, martempering and austempering alter structure/properties through controlled heating/cooling. Failure may be ductile, brittle, notch-sensitive, fatigue-related or creep-related. Testing is destructive or non-destructive according to effect on component.

Study and application method

Describe purpose, basic sequence and resulting property trend. For testing, name the information obtained and give an example.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe purpose, basic sequence and resulting property trend. For testing, name the information obtained and give an example.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Heat-treatment results depend on material and cycle; do not assume one process universally increases every property.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Metrology, Standards, Fits and Form Measurement

Metrology terms and errors; standards/system components; limits/fits/tolerances; vernier/micrometer; geometric form measurements.

15 hours | CO3 | Understand · Apply

Learning goals

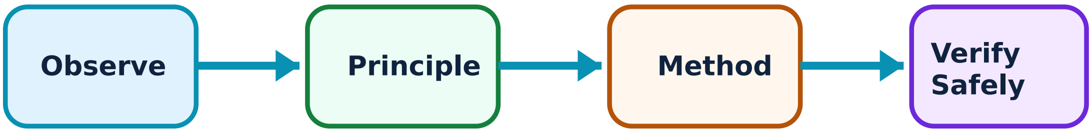
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Metrology, Standards, Fits and Form Measurement: identify the system, state its principle, follow the correct method and verify the result safely.

1. Metrology and measurement quality−

Metrology is science of measurement. Accuracy is closeness to true value; precision describes repeatability. Sensitivity, range and systematic/random errors influence confidence in measurement.

Study and application method

Identify measured quantity, instrument range/least count and likely error source; repeat readings when appropriate.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify measured quantity, instrument range/least count and likely error source; repeat readings when appropriate.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A set of precise readings can still be inaccurate if systematic error is present.

2. Standards, limits, fits and tolerances−

Primary and secondary standards provide traceability. Tolerance, allowance and deviation define permitted dimensional variation. Hole-basis/shaft-basis systems produce clearance, transition or interference fits.

Study and application method

Write basic size, upper/lower limits and relationship between mating parts before classifying fit.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Write basic size, upper/lower limits and relationship between mating parts before classifying fit.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Allowance is not a synonym for tolerance; it is an intentional difference at maximum-material condition.

3. Linear and form measurement−

Vernier calipers and micrometers measure linear features. Form measurement evaluates straightness, flatness, roundness, cylindricity, squareness, parallelism, concentricity and coaxiality.

Study and application method

Choose instrument by feature and required resolution, check zero and align it with measured dimension.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Choose instrument by feature and required resolution, check zero and align it with measured dimension.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not use excessive measuring force or read a scale with parallax.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Angular and Geometric Measurement, Gauges and Comparators

Bevel protractor, sine bar, spirit level, dial gauge, depth gauge, clinometer, autocollimator, angle gauges; GO/NO-GO gauges; comparator types.

15 hours | CO4 | Understand · Apply

Learning goals

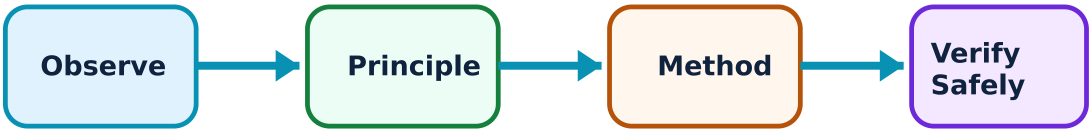
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Angular and Geometric Measurement, Gauges and Comparators: identify the system, state its principle, follow the correct method and verify the result safely.

1. Angular and geometric instruments−

Bevel protractors, sine bars, spirit levels, dial gauges, depth gauges, clinometers, autocollimators and angle gauges support inspection of angles and geometric relationships.

Study and application method

State measurand, setup reference and reading method. Use a labelled setup sketch for sine bar or dial gauge explanations.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State measurand, setup reference and reading method. Use a labelled setup sketch for sine bar or dial gauge explanations.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Instrument reading is valid only when datum surface and alignment are stable/clean.

2. Gauges and GO–NO-GO inspection−

Plug, ring, snap, slip, feeler and thread-pitch gauges check dimensions or attributes efficiently. GO–NO-GO gauges apply limits to accept/reject components without reporting a numerical size.

Study and application method

Identify feature to inspect, select the gauge and state what GO and NO-GO response means.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify feature to inspect, select the gauge and state what GO and NO-GO response means.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: A gauge must not be forced; forcing can damage both gauge and component.

3. Comparators−

Comparators amplify small deviations from a reference. Mechanical, optical, electrical and pneumatic types differ in sensing/amplification principle and application.

Study and application method

Compare principle, sensitivity, environment and typical use in a concise table.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

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Common mistake / precaution: Comparator gives deviation relative to standard; it must be set against a reference first.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Basic Engineering Materials and Structure

Use the concept flow to revise observation, principle, method and verification.

Module 2: Properties, Heat Treatment and Material Testing

Use the concept flow to revise observation, principle, method and verification.

Module 3: Metrology, Standards, Fits and Form Measurement

Use the concept flow to revise observation, principle, method and verification.

Module 4: Angular and Geometric Measurement, Gauges and Comparators

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Prepare material-classification, crystal-system and alloy application notes.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare properties, heat treatments, failures and DT/NDT examples.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Study metrology terms, fits/tolerances and instrument comparisons.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Prepare diagrams/charts for angular instruments, gauges and comparator types.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Basic Engineering Materials and Structure

Explain Material classification and atomic bonding and state one correct method, application or precaution.−

Engineering materials are selected by structure, properties, processing and use. Atomic structure and electronic configuration influence bonding. Metallic, ionic and covalent bonding provide different combinations of conductivity, strength and ductility.

Answer framework: Classify a material first, then connect atomic/bonding description to one observable property and application.

Important point: Material class does not by itself guarantee performance; service condition and processing also matter.

Explain Crystallography and packing and state one correct method, application or precaution.−

Crystalline metals have ordered atomic arrangement. Crystal systems, unit-cell parameters, packing factor and BCC/FCC/HCP structures help describe how atoms occupy space and relate structure to properties.

Answer framework: Use a labelled unit-cell sketch and state coordination/packing idea at the required level before comparing structures.

Important point: Do not confuse a grain in a real metal with a single unit cell.

Explain Ferrous, non-ferrous and special materials and state one correct method, application or precaution.−

Steel, stainless steel and cast iron are official ferrous examples. Aluminium/copper alloys, magnesium and titanium alloys are non-ferrous examples. Smart materials and composites are special materials selected for specialised response or combined properties.

Answer framework: Compare composition category, key property and typical engineering use in a table.

Important point: Alloys of a base metal can differ significantly from pure metal; avoid generalising one property.

Module 2 — Properties, Heat Treatment and Material Testing

Explain Mechanical and thermal properties and state one correct method, application or precaution.−

Elasticity, plasticity, yield, tensile behaviour, ductility, toughness, hardness, fatigue and creep describe mechanical response. Thermal conductivity and expansion describe heat-related response relevant to service and dimensional change.

Answer framework: Define each property, state a simple test/condition or engineering example, and distinguish strength from hardness/toughness.

Important point: Hardness, strength and toughness are related but not interchangeable terms.

Explain Magnetic and electrical properties and state one correct method, application or precaution.−

Permeability, retentivity and coercivity describe magnetic behaviour. Conductivity and resistivity describe electrical transport. Selection depends on whether a component must conduct, resist current, store flux or avoid magnetisation.

Answer framework: State unit/meaning where applicable and link property to a component such as conductor, resistor, core or sensor.

Important point: High electrical conductivity does not imply high magnetic permeability.

Explain Heat treatment, failures and testing and state one correct method, application or precaution.−

Annealing, normalizing, hardening, tempering, case hardening, martempering and austempering alter structure/properties through controlled heating/cooling. Failure may be ductile, brittle, notch-sensitive, fatigue-related or creep-related. Testing is destructive or non-destructive according to effect on component.

Answer framework: Describe purpose, basic sequence and resulting property trend. For testing, name the information obtained and give an example.

Important point: Heat-treatment results depend on material and cycle; do not assume one process universally increases every property.

Module 3 — Metrology, Standards, Fits and Form Measurement

Explain Metrology and measurement quality and state one correct method, application or precaution.−

Metrology is science of measurement. Accuracy is closeness to true value; precision describes repeatability. Sensitivity, range and systematic/random errors influence confidence in measurement.

Answer framework: Identify measured quantity, instrument range/least count and likely error source; repeat readings when appropriate.

Important point: A set of precise readings can still be inaccurate if systematic error is present.

Explain Standards, limits, fits and tolerances and state one correct method, application or precaution.−

Primary and secondary standards provide traceability. Tolerance, allowance and deviation define permitted dimensional variation. Hole-basis/shaft-basis systems produce clearance, transition or interference fits.

Answer framework: Write basic size, upper/lower limits and relationship between mating parts before classifying fit.

Important point: Allowance is not a synonym for tolerance; it is an intentional difference at maximum-material condition.

Explain Linear and form measurement and state one correct method, application or precaution.−

Vernier calipers and micrometers measure linear features. Form measurement evaluates straightness, flatness, roundness, cylindricity, squareness, parallelism, concentricity and coaxiality.

Answer framework: Choose instrument by feature and required resolution, check zero and align it with measured dimension.

Important point: Do not use excessive measuring force or read a scale with parallax.

Module 4 — Angular and Geometric Measurement, Gauges and Comparators

Explain Angular and geometric instruments and state one correct method, application or precaution.−

Bevel protractors, sine bars, spirit levels, dial gauges, depth gauges, clinometers, autocollimators and angle gauges support inspection of angles and geometric relationships.

Answer framework: State measurand, setup reference and reading method. Use a labelled setup sketch for sine bar or dial gauge explanations.

Important point: Instrument reading is valid only when datum surface and alignment are stable/clean.

Explain Gauges and GO–NO-GO inspection and state one correct method, application or precaution.−

Plug, ring, snap, slip, feeler and thread-pitch gauges check dimensions or attributes efficiently. GO–NO-GO gauges apply limits to accept/reject components without reporting a numerical size.

Answer framework: Identify feature to inspect, select the gauge and state what GO and NO-GO response means.

Important point: A gauge must not be forced; forcing can damage both gauge and component.

Explain Comparators and state one correct method, application or precaution. −

Comparators amplify small deviations from a reference. Mechanical, optical, electrical and pneumatic types differ in sensing/amplification principle and application.

Answer framework: Compare principle, sensitivity, environment and typical use in a concise table.

Important point: Comparator gives deviation relative to standard; it must be set against a reference first.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Basic Engineering Materials and Structure.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Properties, Heat Treatment and Material Testing.

4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Metrology, Standards, Fits and Form Measurement.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Angular and Geometric Measurement, Gauges and Comparators.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Material classification; atomic structure/bonding; crystallography; ferrous/non-ferrous alloys; smart materials and composites..
2. Develop a complete answer or practical plan for: Mechanical, thermal, magnetic and electrical properties; heat treatments; metal failure; destructive and non-destructive testing..
3. Develop a complete answer or practical plan for: Metrology terms and errors; standards/system components; limits/fits/tolerances; vernier/micrometer; geometric form measurements..
4. Develop a complete answer or practical plan for: Bevel protractor, sine bar, spirit level, dial gauge, depth gauge, clinometer, autocollimator, angle gauges; GO/NO-GO gauges; comparator types..

LAST-MILE REVISION

Quick Revision

Module 1: Basic Engineering Materials and Structure

Material classification; atomic structure/bonding; crystallography; ferrous/non-ferrous alloys; smart materials and composites.

- Match each topic to CO1.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Module 2: Properties, Heat Treatment and Material Testing

Mechanical, thermal, magnetic and electrical properties; heat treatments; metal failure; destructive and non-destructive testing.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Metrology, Standards, Fits and Form Measurement

Metrology terms and errors; standards/system components; limits/fits/tolerances; vernier/micrometer; geometric form measurements.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Angular and Geometric Measurement, Gauges and Comparators

Bevel protractor, sine bar, spirit level, dial gauge, depth gauge, clinometer, autocollimator, angle gauges; GO/NO-GO gauges; comparator types.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Material classification and atomic bonding	Engineering materials are selected by structure, properties, processing and use.
Crystallography and packing	Crystalline metals have ordered atomic arrangement.
Ferrous, non-ferrous and special materials	Steel, stainless steel and cast iron are official ferrous examples.
Mechanical and thermal properties	Elasticity, plasticity, yield, tensile behaviour, ductility, toughness, hardness, fatigue and creep describe mechanical response.
Magnetic and electrical properties	Permeability, retentivity and coercivity describe magnetic behaviour.
Heat treatment, failures and testing	Annealing, normalizing, hardening, tempering, case hardening, martempering and austempering alter structure/properties through controlled heating/cooling.
Metrology and measurement quality	Metrology is science of measurement.
Standards, limits, fits and tolerances	Primary and secondary standards provide traceability.
Linear and form measurement	Vernier calipers and micrometers measure linear features.
Angular and geometric instruments	Bevel protractors, sine bars, spirit levels, dial gauges, depth gauges, clinometers, autocollimators and angle gauges support inspection of angles and geometric relationships.
Gauges and GO–NO-GO inspection	Plug, ring, snap, slip, feeler and thread-pitch gauges check dimensions or attributes efficiently.
Comparators	Comparators amplify small deviations from a reference.

LEARNING RESOURCES

References

Officially listed print resources

1. V. Raghavan, Material Science and Engineering.
2. R. K. Rajput, Engineering Materials and Metallurgy.
3. R. K. Jain, Engineering Metrology.
4. Raghavendra and Krishnamurthy, Engineering Metrology and Measurements.

Officially listed web resources

- https://nptel.ac.in/courses/112108150
- https://nptel.ac.in/courses/112104250

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